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Lightweight Post-training for Controllable Vietnamese Summarization



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1. Introduction
2. Related works
3. Propose method
4. Experiments
5. Conclusion





1. Introduction

2. Related works

3. Propose method

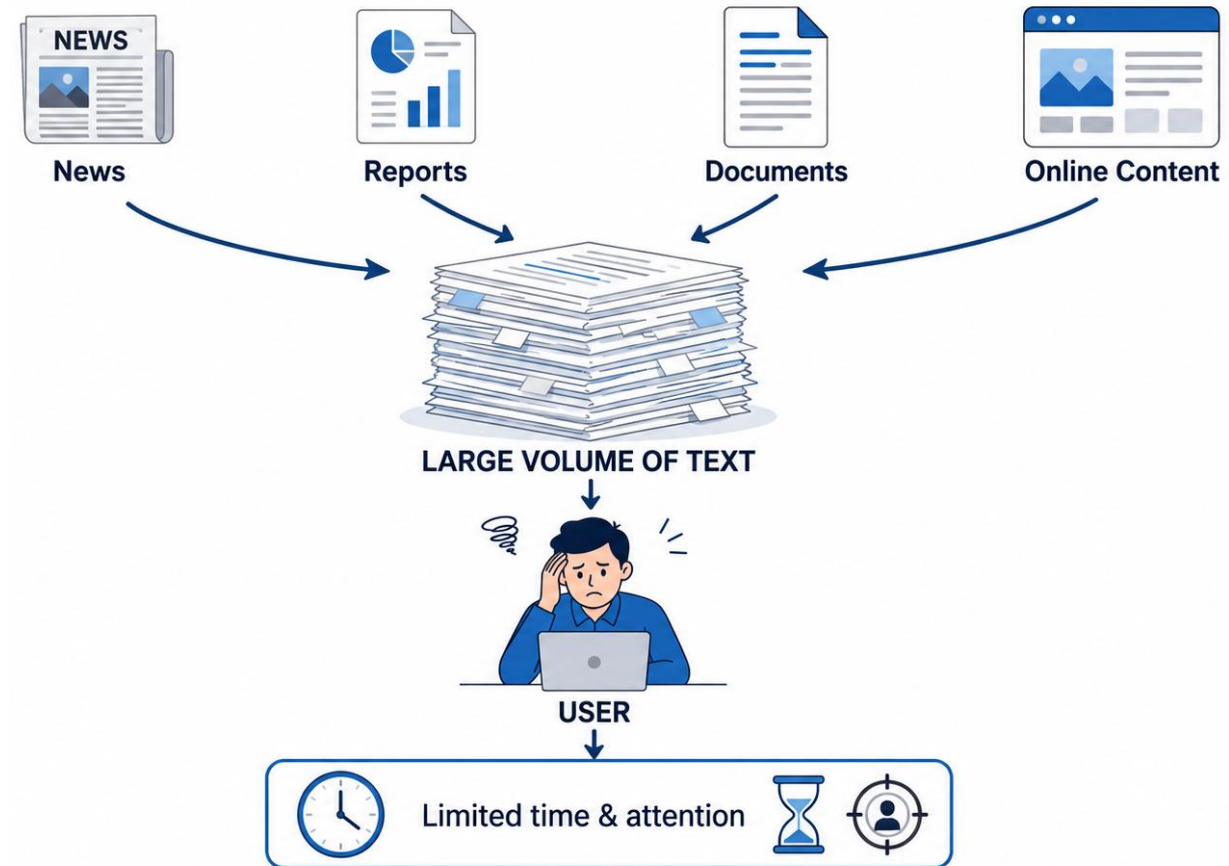
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5. Conclusion



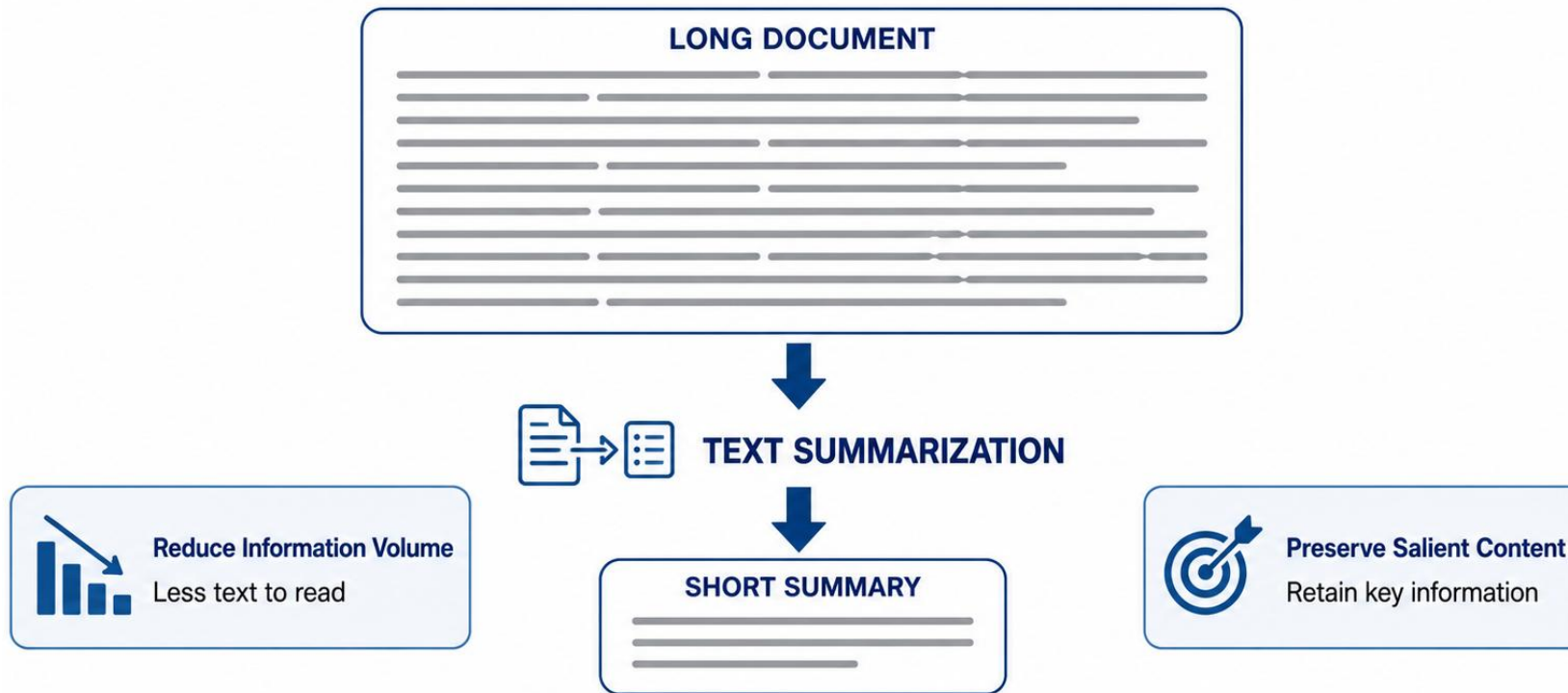
Growing volume of textual information

- AI-driven content creation is rapidly increasing the volume of textual information
 - Text is continuously generated across news, reports, documents, and online platforms.
- ➔ Too much information, but limited time and attention
- ➔ **Growing need for text summarization**



Documentation summarization

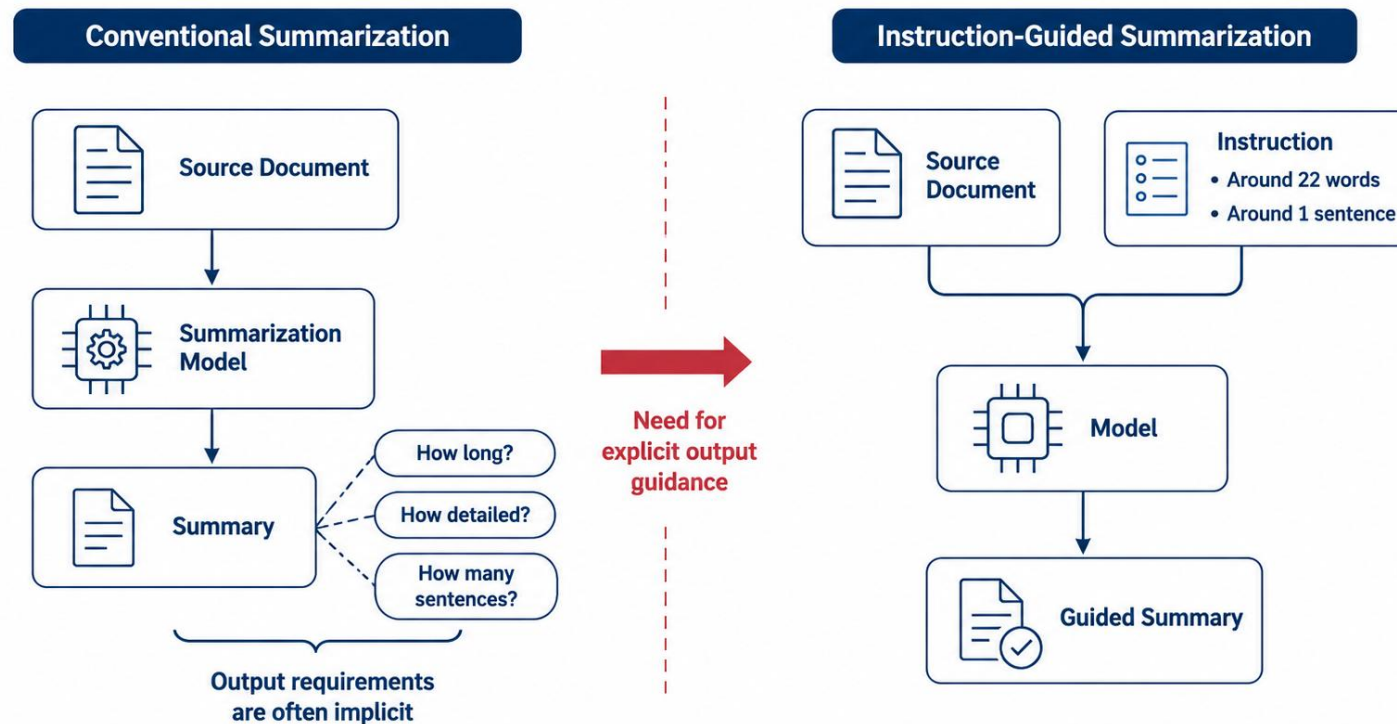
- Text summarization transforms a source document into a shorter representation of its main content.
- It identifies and condenses the most relevant information from the original text.



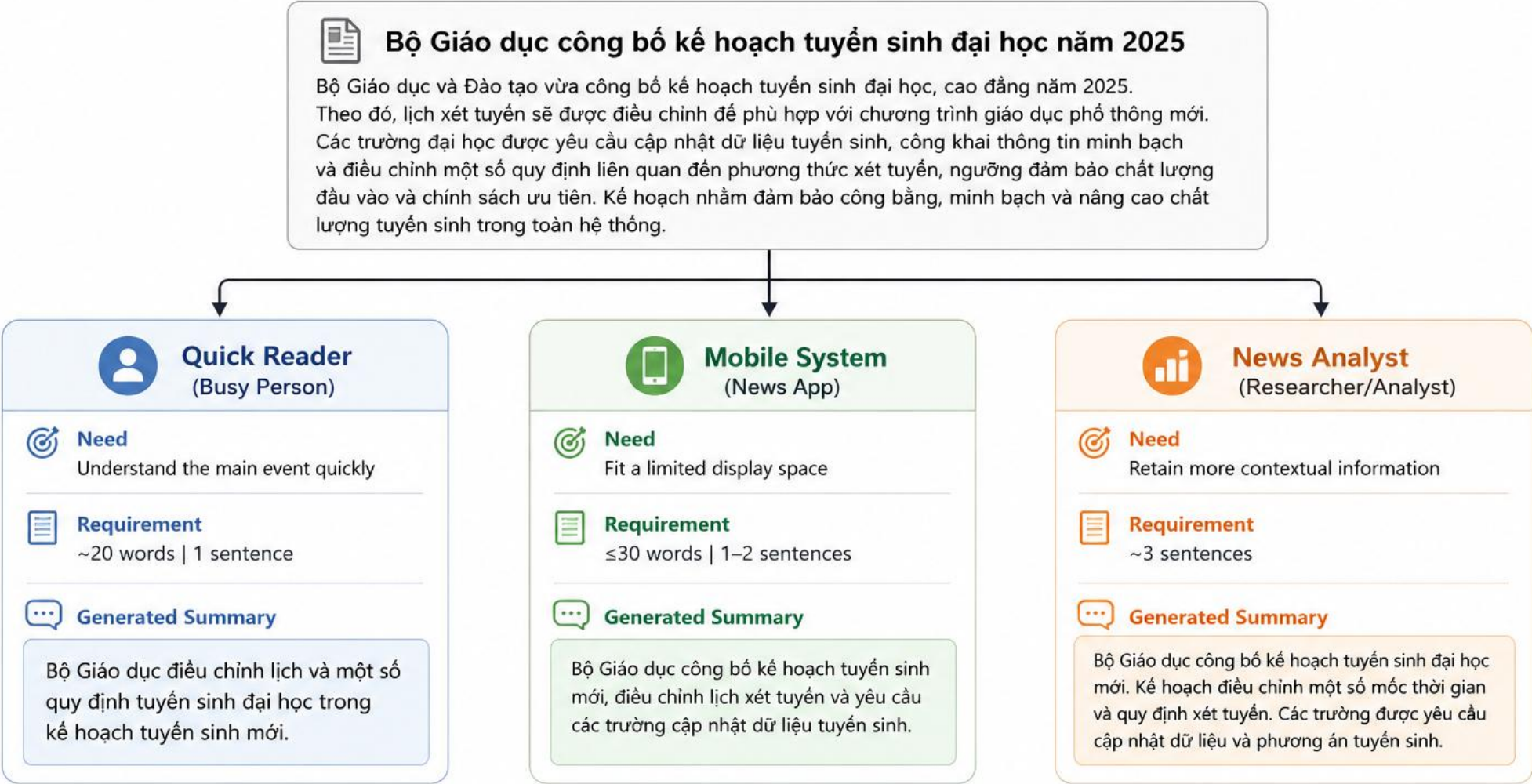
From summarization to instruction-guided summarization

Traditional summarization mainly focuses on generating a concise and informative summary.

However, the desired output characteristics are often left implicit.



A Practical Example



Length control vietnamese summarization task

An instance:

$$(x, c, y^*)$$

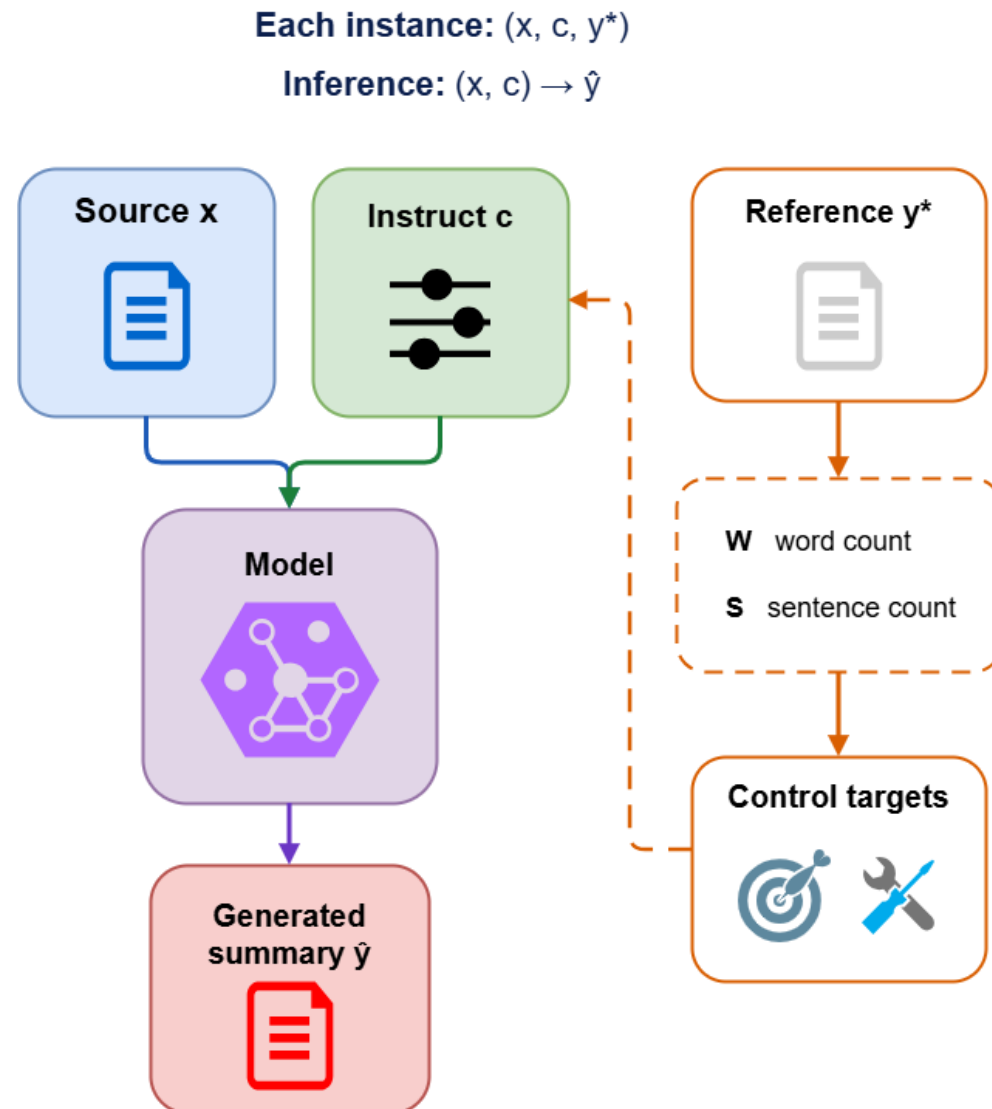
- x : vietnamese documentation
- c : target instructions/controls
- y^* : reference summary

Inference:

$$(x, c) \rightarrow \hat{y}$$

Goal:

- Preserve importance contents
- Follow the length budget





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2. Related works

3. Propose method

4. Experiments

5. Conclusion



Vietnamese Text Summarization

Vietnamese summarization has been studied for news articles, instructional texts, and multi-document collections.

Benchmark	Year	Description
VNDS [1]	2019	Single-document news summarization from Vietnamese articles
ViMs [2]	2020	Multi-document news summarization from clusters of related articles
WikiLingua [3]	2020	Single-document instructional summarization in a multilingual setting
VLSP AbMuSu [4]	2023	Multi-document news summarization benchmark and shared task

Recent work has further improved Vietnamese abstractive summarization through pretrained text-to-text models such as ViT5 [5] and BRIO-style training [6].

[1] Nguyen, Van-Hau, et al. "Vnds: A vietnamese dataset for summarization." 2019 6th NAFOSTED conference on information and computer science (NICS). IEEE, 2019.

[2] Tran, Nhi-Thao, et al. "ViMs: a high-quality Vietnamese dataset for abstractive multi-document summarization." Language Resources and Evaluation 54.4 (2020): 893-920..

[3] Ladhak, Faisal, et al. "WikiLingua: A new benchmark dataset for cross-lingual abstractive summarization." Findings of the Association for Computational Linguistics: EMNLP 2020. 2020.

[4] Tran, Mai-Vu, et al. "Overview of the VLSP 2022--Abmusu Shared Task: A Data Challenge for Vietnamese Abstractive Multi-document Summarization." arXiv preprint arXiv:2311.15525 (2023).

[5] Phan, Long, et al. "ViT5: Pretrained text-to-text transformer for Vietnamese language generation." Proceedings of the 2022 conference of the North American chapter of the association for computational linguistics: human language technologies: student research workshop. 2022.

[6] Lam, Khang, et al. "Abstractive text summarization using the BRIO training paradigm." Findings of the Association for Computational Linguistics: ACL 2023. 2023..

Controllable and Length-Aware Summarization

Prior work has explored several ways to explicitly control summary characteristics.

Category	Publications	Description
Architecture-based length control	[7], [8]	Controls output length through decoder conditioning or positional representations
General controllable summarization	[9], [10]	Uses explicit signals such as length, style, or keywords
Precise length control	[11]	Targets exact or tightly constrained output length
Instruction-based control	[12]–[14]	Expresses output requirements through natural-language instructions

[7] Y. Kikuchi, G. Neubig, R. Sasano, H. Takamura, and M. Okumura, “Controlling Output Length in Neural Encoder-Decoders,” in *Proc. EMNLP*, 2016, pp. 1328–1338.

[8] S. Takase and N. Okazaki, “Positional Encoding to Control Output Sequence Length,” in *Proc. NAACL-HLT*, 2019, pp. 3999–4004.

[9] A. Fan, D. Grangier, and M. Auli, “Controllable Abstractive Summarization,” in *Proc. 2nd Workshop on Neural Machine Translation and Generation*, 2018, pp. 45–54.

[10] J. He, W. Kryscinski, B. McCann, N. Rajani, and C. Xiong, “CTRLsum: Towards Generic Controllable Text Summarization,” in *Proc. EMNLP*, 2022, pp. 5879–5915.

[11] L. Miculicich, Y. Xie, S. Wang, and P. He, “Summarization with Precise Length Control,” *arXiv preprint arXiv:2305.05171*, 2023.

[12] Y. Liu, A. Fabbri, J. Chen, Y. Zhao, S. Han, S. Joty, P. Liu, D. Radev, C.-S. Wu, and A. Cohan, “Benchmarking Generation and Evaluation Capabilities of Large Language Models for Instruction Controllable Summarization,” in *Findings of NAACL*, 2024, pp. 4481–4501.

[13] Jie et al., “Prompt-Based Length Controlled Generation with Multiple Control Types,” *Findings of ACL*, 2024.

[14] Juseon-Do et al., “InstructCMP: Length Control in Sentence Compression through Instruction-based Large Language Models,” *Findings of ACL*, 2024.

Lightweight Post-training and Reward Optimization

Recent post-training methods enable efficient task adaptation and reward-based refinement of generation behavior.

Category	Publications	Description
Parameter-efficient adaptation	[15]	Learns low-rank adapters while freezing the pretrained backbone
Reward-based summarization	[16]; [17]	Refines summary generation using learned or task-specific rewards
Policy optimization	[18]	Uses group-relative policy optimization for efficient reward-based alignment
Reward optimization risks	[19]; [20]; [21]	Proxy rewards may be optimized without corresponding task-quality gains

[15] Hu et al., “LoRA: Low-Rank Adaptation of Large Language Models,” ICLR, 2022.

[16] Böhm et al., “Better Rewards Yield Better Summaries: Learning to Summarise Without References,” EMNLP-IJCNLP, 2019.

[17] Stiennon et al., “Learning to Summarize from Human Feedback,” NeurIPS, 2020.

[18] Shao et al., “DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models,” arXiv:2402.03300, 2024.

[19] Amodei et al., “Concrete Problems in AI Safety,” arXiv:1606.06565, 2016.

[20] Skalse et al., “Defining and Characterizing Reward Hacking,” NeurIPS, 2022.

[21] Gao et al., “Scaling Laws for Reward Model Overoptimization,” ICML, 2023.

Research Gap and Method Positioning

Observations from Prior Work	Remaining Need	Project Positioning
Vietnamese summarization mainly targets general summary quality [1–6]	Explicit control of word and sentence budgets	Controllable Vietnamese summarization under explicit output budgets
Control has shifted toward natural-language instructions , but precise adherence remains difficult [7–14]	Preserve content while satisfying multiple explicit constraints	Joint content-and-control optimization
Reward-based post-training can refine behavior, but proxy rewards may induce failure-prone optimization [15–21]	Efficient adaptation with robust reward optimization	Lightweight post-training with reward safeguards

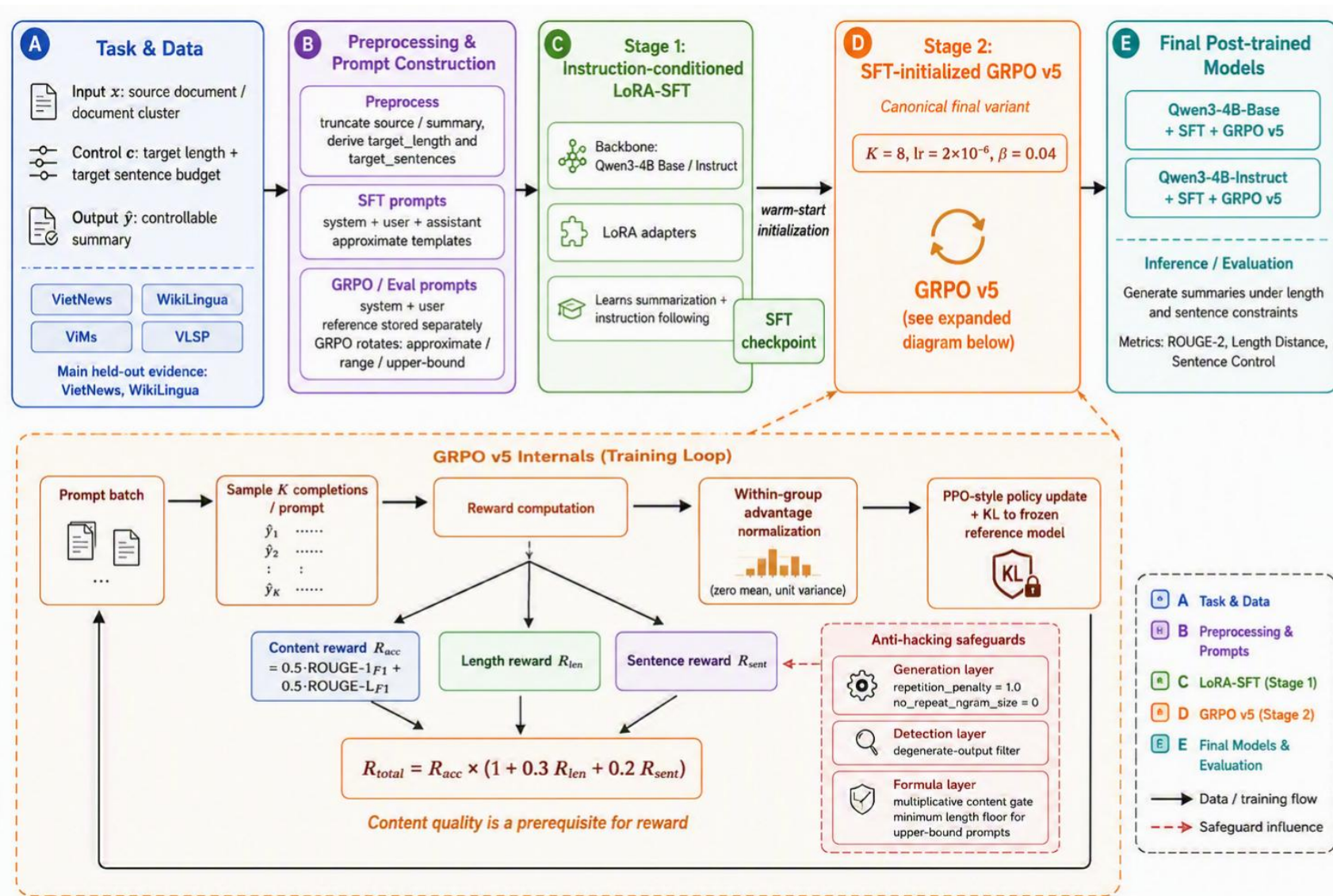
This project studies lightweight post-training for controllable Vietnamese summarization under length constraints.



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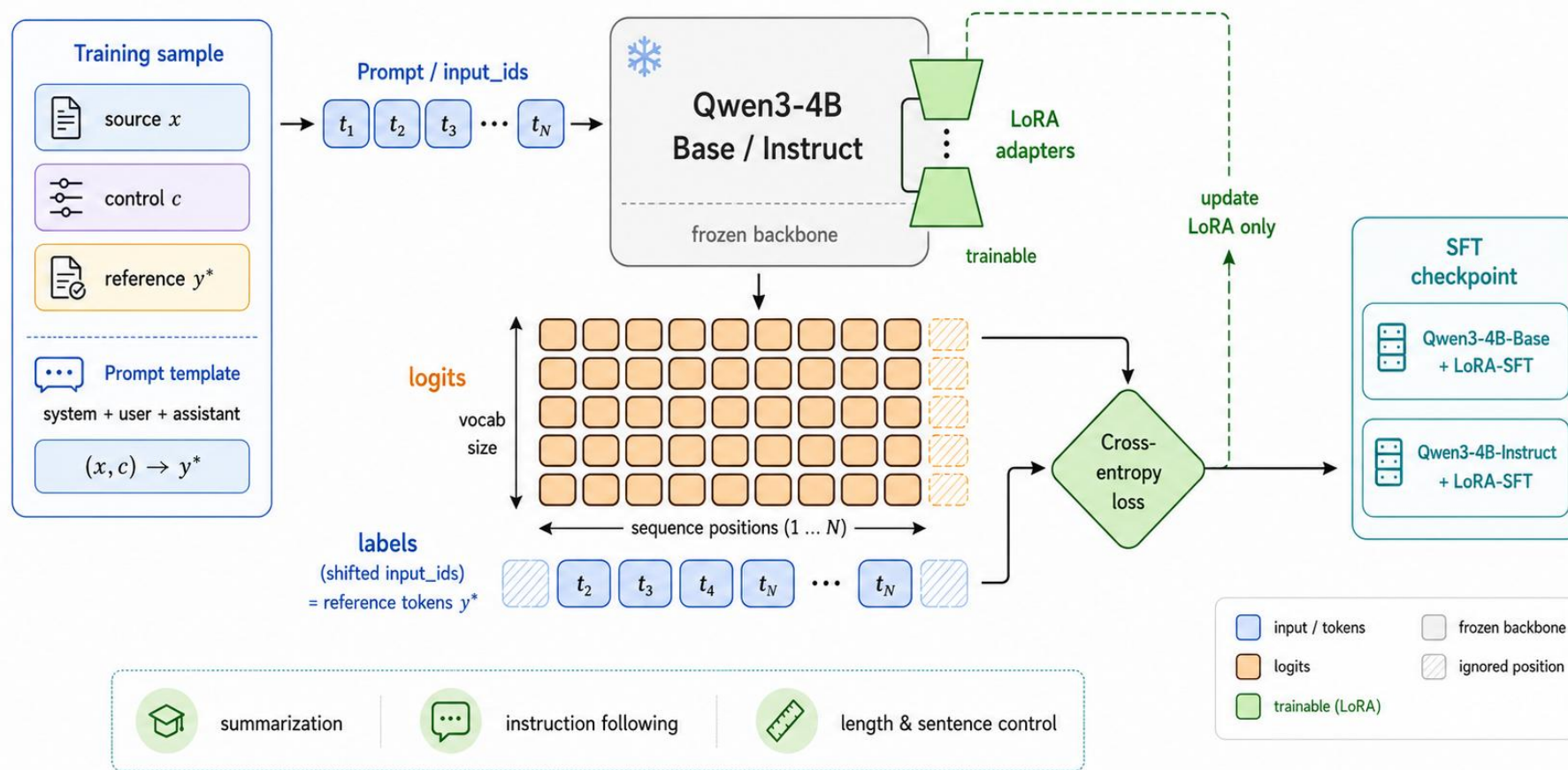


General frameworks

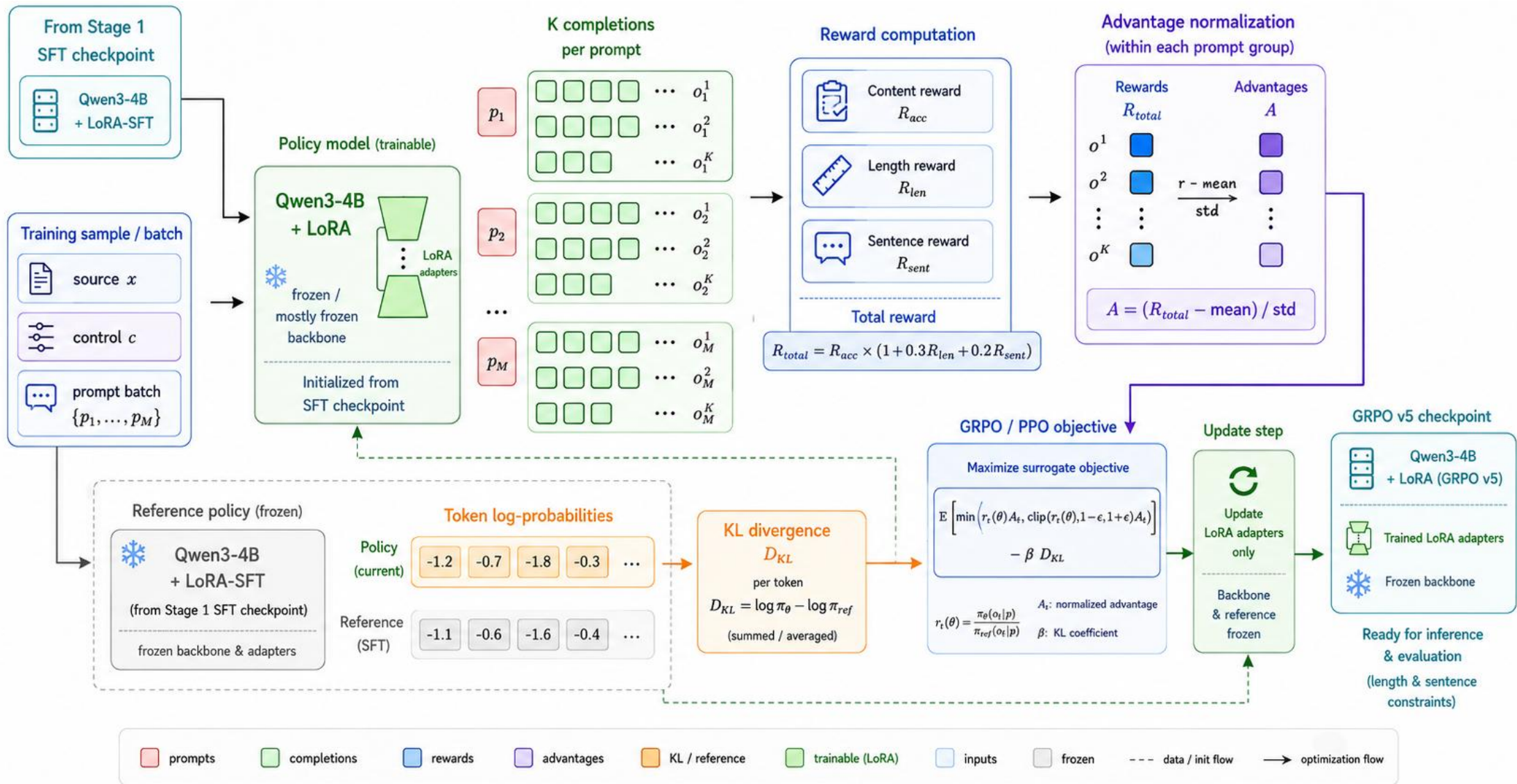


Stage 1: Instruction-conditioned LoRA-SFT

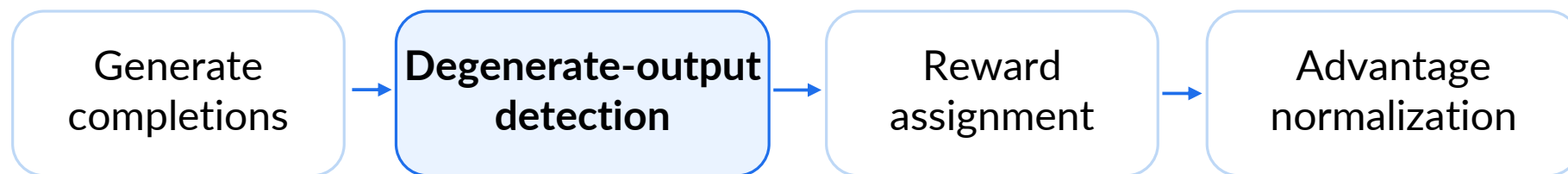
The pretrained model is not yet specialized for controllable Vietnamese summarization
→ SFT first establishes summarization and instruction-following capability.



Stage 2: GRPO Alignment



Anti-hacking safeguards



Each completion is checked before assigning control rewards.

Detection rule	Threshold / signal
Empty output	Empty or whitespace only
Long token blob	Single token without spaces (>25) chars
Too few alphabetic chars	Alphabetic characters (<3)
Low alphabetic ratio	Alphabetic ratio (<12%)
Dominant-token repetition	One token (>60%)
Low token diversity	Unique tokens (<30%)

If flagged:

$$R_{\text{len}} = R_{\text{sent}} = 0$$

➔ **Degenerate completions cannot earn control bonuses.**

Reward design

Combine reward:

$$R_{total} = R_{acc}(1 + 0.3 R_{len} + 0.2 R_{sent})$$

Reward	Formula	Meaning
Content quality	$R_{acc} = 0.5 \text{ROUGE-1}_{F1} + 0.5 \text{ROUGE-L}_{F1}$	Measures content overlap and preservation against the reference summary
Length adherence	$R_{len} = \text{Score}(w(\hat{y}), c_{len}) \in [0,1]$	Measures whether the generated word count satisfies the requested length constraint
Sentence adherence	$R_{sent} = \text{Score}(s(\hat{y}), c_{sent}) \in [0,1]$	Measures whether the generated sentence count satisfies the requested sentence constraint

Note: $w()$, $s()$ are word count and sentence count function respectively

➔ Content quality acts as the reward gate, while length and sentence adherence provide only conditional bonuses.



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3. Propose method
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Experimental Questions

Research questions	Detail
RQ1 – Overall effectiveness	Does the final pipeline improve both content quality and control adherence over the raw backbones?
RQ2 – Stage contribution	Which training stage contributes the largest performance gain?
RQ3 – GRPO initialization	Does GRPO require an SFT warm start?
RQ4 – Reward robustness	Can the safeguard package mitigate reward-hacking failures?

➔ Each research question is answered by a dedicated experiment.

Datasets and Evaluation Boundary

Dataset overview

Dataset	Domain	Characteristic	Training	Validation	Test
VietNews	Vietnamese news	Single-document	105,418	22,642	2,000
WikiLingua	How-to articles	Single-document	13,999	1,680	500
ViMs	Vietnamese news clusters	Multi-document	240	60	300*
VLSP	Vietnamese news clusters	Multi-document	285	15	300*

Pipeline usage

- SFT filters short VietNews titles and excludes VLSP, while GRPO uses all VietNews training items and includes VLSP.

SFT	GRPO	Evaluation
VietNews: 96,911	VietNews: 105,418	VietNews: 2,000
WikiLingua: 13,999	WikiLingua: 13,999	WikiLingua: 500
ViMs: 240	ViMs: 240	ViMs: 300*
VLSP: Not used	VLSP: 285	VLSP: 300*
Total: 111,150	Total: 119,942	Total: 3,100



(*) ViMs and VLSP test items overlap with the project training and validation pools.

Evaluation Metrics

Metric	Definition	Meaning	Interpretation
ROUGE-2 F1	$F_1^{(2)} = \frac{2P^{(2)}R^{(2)}}{P^{(2)} + R^{(2)}}$	Measures overlapping bigrams (consecutive word pairs) between the generated and gold standard summaries, balancing precision and recall.	Higher is better
Absolute Length Distance	$LenDist = N_{gen} - N_{target} $	Measures the absolute word-count difference between the model's generated text and the strict length limit requested in the instruction	Lower is better

$$P^{(2)} = \frac{\text{matched bigrams}}{\text{generated bigrams}}, R^{(2)} = \frac{\text{matched bigrams}}{\text{reference bigrams}}$$

N_{gen} : number of words in the generated summary

N_{target} : target word count specified by the control instruction

Evaluation Protocol and Baselines

Group	Models
Project baselines	Qwen3-4B-Base, Qwen3-4B-Instruct
Final project systems	Base + SFT + GRPO v5, Instruct + SFT + GRPO v5
External baselines	VietAI, GPT-4o, GPT-3.5-Turbo, Qwen3-14B, Llama3.3-70B, Phi4-14B, Sailor-20B-chat, VinBigdata-7B

Frozen Evaluation Protocol

Evaluation Baselines

Setting	Configuration
Decoding	Temperature 0.3, Top-p 0.9
Sampling	Random sampling
Max new tokens	256
Batch size	8
Context length	3,072 tokens
Prompt policy	Fixed approximate template

RQ1 – Overall effectiveness

Model	VietNews	WikiLingua	ViMs	VLSP
<i>Project systems</i>				
Qwen3-4B-Base + SFT + GRPO v5	1.16	8.35	89.23	100.50
Qwen3-4B-Instruct + SFT + GRPO v5	<u>1.52</u>	<u>10.25</u>	84.45	82.71
<i>Raw Qwen3 baselines</i>				
Qwen3-4B-Base (raw)	33.03	61.59	92.01	120.33
Qwen3-4B-Instruct (raw)	4.01	23.42	<u>59.80</u>	70.17
<i>External baselines</i>				
VietAI	–	–	–	–
GPT-4o	8	11	187	58
GPT-3.5-Turbo	18	17	47	30
Qwen3-14B	50	45	131	<u>34</u>
Llama3.3-70B-Instruct	14	13	186	73
Phi4-14B	205	214	166	134
Sailor-20B-chat	38	43	268	41
VinBigdata (7B)	115	89	98	82

Table. Absolute summary length distance across the four report datasets.

RQ1 – Overall effectiveness

Model	VietNews	WikiLingua	ViMs	VLSP
<i>Project systems</i>				
Qwen3-4B-Base + SFT + GRPO v5	28.97	<u>26.35</u>	18.16	17.46
Qwen3-4B-Instruct + SFT + GRPO v5	<u>29.62</u>	25.59	21.27	24.39
<i>Raw Qwen3 baselines (frozen project protocol)</i>				
Qwen3-4B-Base (raw)	12.13	3.18	13.46	9.49
Qwen3-4B-Instruct (raw)	16.48	7.44	17.83	15.51
<i>Historical external baselines</i>				
VietAI	34.24	33.12	–	–
GPT-4o	21.61	20.65	<u>44.26</u>	<u>43.37</u>
GPT-3.5-Turbo	28.13	21.09	19.28	35.79
Qwen3-14B	18.51	20.24	44.38	44.27
Llama3.3-70B-Instruct	22.17	19.40	37.54	39.02
Phi4-14B	13.17	14.28	41.85	42.31
Sailor-20B-chat	17.70	18.74	39.47	40.96
VinBigdata (7B)	20.59	21.02	37.98	40.23

Table. ROUGE-2 (%) of models across the four report datasets.

RQ1 – Overall effectiveness

Baseline-Relative Improvement of Final Project Systems

Qwen3-4B-Base + SFT + GRPO v5



Qwen3-4B-Instruct + SFT + GRPO v5

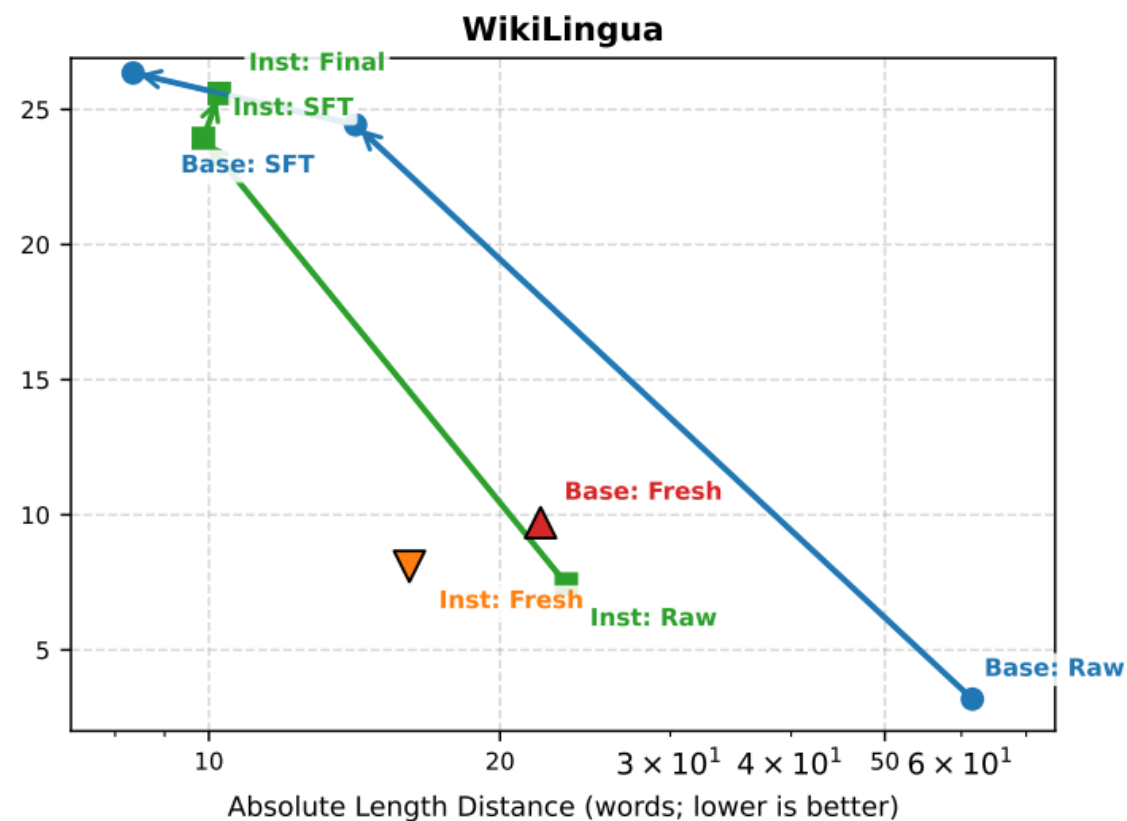
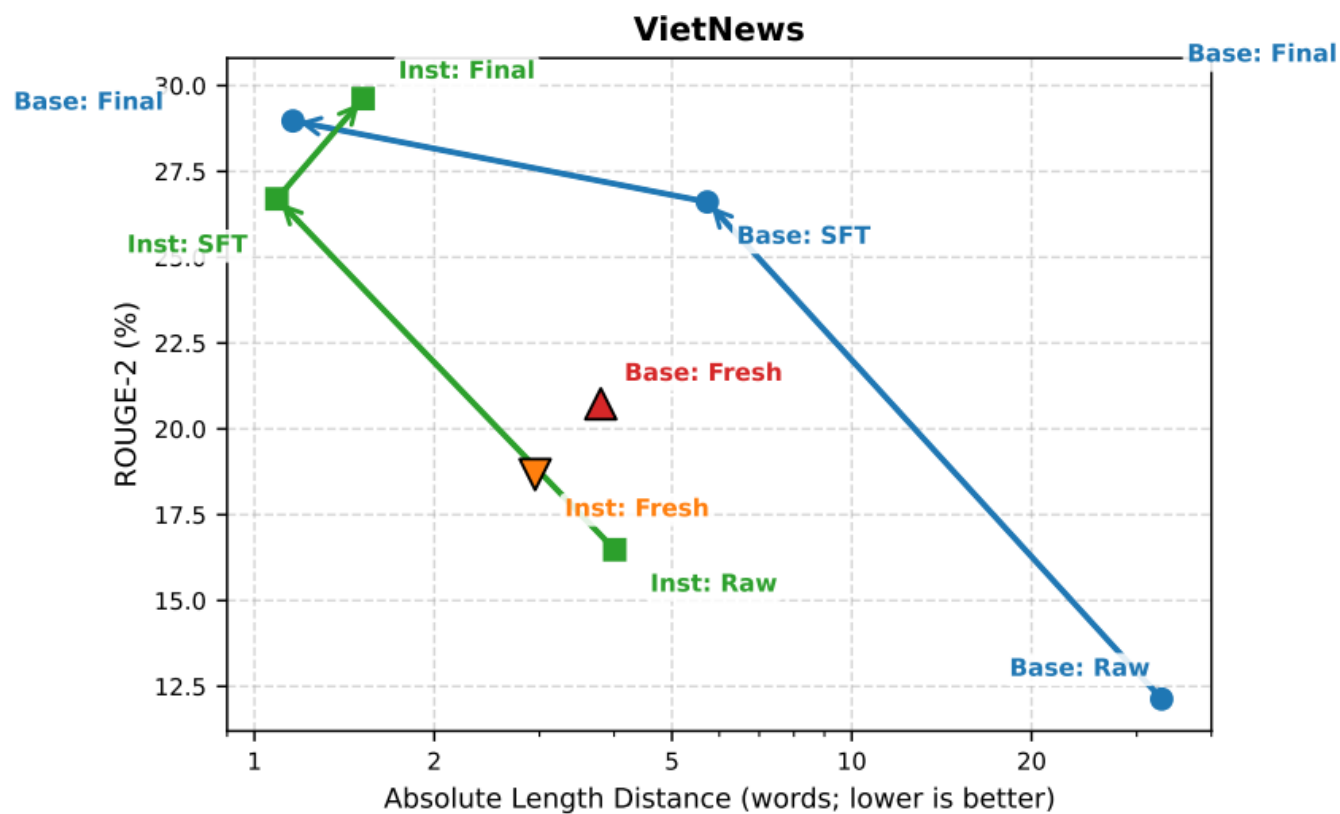


RQ1 – Overall effectiveness

- Both final systems consistently outperform their raw backbones on VietNews and WikiLingua.
- Post-training improves both content quality and length control.
- Performance remains limited on ViMs and VLSP due to predominantly single-document training data.
- The Instruct branch favors content quality, while the Base branch provides stronger length control.
- The final models outperform larger LLM baselines in length control, supporting the effectiveness of the proposed approach

The final pipeline jointly improves content quality and length control over the raw backbones on the main single-document datasets, while multi-document performance remains limited.

RQ2 – Stage contribution



Insight 1 – Largest shift occurs after SFT. SFT phase produces the dominant gain

Insight 2 – GRPO provides further refinement. ROUGE-2 improves for both backbones on both datasets.

Insight 3 – Control refinement depends on the backbone. GRPO does not uniformly improve Length Distance

SFT contributes the largest performance gain by establishing the core summarization and instruction-following capability.

RQ3 – GRPO initialization

Backbone	Fresh GRPO	SFT-init GRPO	Difference
Base	20.73	28.97	+8.24
Instruct	18.67	29.62	+10.95
Base	9.71	26.35	+16.64
Instruct	8.09	25.59	+17.50

**Compare between GRPO only
and SFT init GRPO performance
by ROUGE-2**

**Compare between GRPO only
and SFT init GRPO performance
by Len-Dist**

Dataset	Backbone	Fresh	SFT-init
VietNews	Base	3.80	1.16
VietNews	Instruct	2.95	1.52
WikiLingua	Base	22.03	8.35
WikiLingua	Instruct	16.12	10.25

GRPO benefits from an SFT warm start. Fresh GRPO is substantially weaker and does not replace the SFT stage.

RQ4 – Reward robustness

Case	Reference	v3 output (failure)	v5 output (recovered)
Base fresh idx 14	Điều tra vụ anh rể dùng dao đâm chết em vợ vào rạng sáng	Output <i>assistant assistant assistant assistant assistant assistant ... (×128 tokens)</i> Failure signature • Pure repetition loop — no content • ROUGE-2 = 0.00 • LenErr = 814.29% • R_acc = 0.00	Output <i>Anh C. bị anh rể đâm chết tại nhà.</i> Recovered properties • Short, content-bearing headline • Named entity retained • Budget restored • ROUGE-2 = 0.19
Base SFT-init idx 7	5 năm chưa xử xong vụ án “lạ” có hơn 30 luật sư bào chữa miễn phí	Output <i>Bị cáo Tuyết bị cấp sơ thẩm 2 lần tuyên phạt 12 năm tù . Bị cáo Tuyết kháng cáo , nên 2 lần vụ án được đưa ra xét xử phúc thẩm và đều có chung 1 kết quả , huỷ án</i> Failure signature • 245 words — punctuation-heavy blow-up • Meaningful start → punctuation flood • LenErr = 1189.47% • ROUGE-2 = 0.034	Output <i>Vụ án “lạ” : Huỷ án 2 lần , bị cáo vẫn bị tạm giam 5 năm !</i> Recovered properties • Compact, accurate rewrite • Budget restored • Content preserved • ROUGE-2 = 0.22

RQ4 – Reward robustness

Branch	Metric	v3	v5	Change
Base Fresh	Degenerate rate ↓	23.81%	0.32%	≈74× lower
	$R_{acc} = 0\% \downarrow$	27.16%	1.94%	-25.22 pts
	ROUGE-2 ↑	10.58	18.66	+8.08
Base SFT-init	LenErr ↓	425.00%	14.94%	-410.06 pts
	ROUGE-2 ↑	18.03	26.38	+8.35

- Degenerate outputs are almost eliminated.
- ROUGE-2 improves after applying the safeguards.
- The largest improvements are observed in the unstable Base branches.

The safeguard package sharply mitigates the observed reward-hacking failures and removes catastrophic degeneration in the unstable branches.



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Key Conclusions:

- Studied Vietnamese controllable summarization under explicit length constraints.
- Developed a lightweight two-stage post-training recipe combining LoRA-SFT and SFT-initialized GRPO.
- Demonstrated clear gains in content quality and length control over the raw backbones on the main single-document datasets.
- Benchmarked the final systems against representative baselines.

Future works:

- Improve the performance of the method on multi-doc summary datasets.

A graphic on the left side of the slide. It features a dark blue background with a large, stylized circular pattern of red dots. The dots are arranged in concentric, slightly irregular rings, creating a sense of depth and movement. In the center of this pattern, the word "HUST" is written in a bold, white, sans-serif font.

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THANK YOU !